

of cybermemorials found a word count similar to the ones we report here.

Understandably, ED comments express less positive emotion than non-ED post-mortem comments. That ED comments include significantly less positive emotion words than non-ED comments may suggest that those expressing ED are more likely to be in a depressed state. This is consistent with previous research that looked at linguistic signs of depression (Pennebaker, Mehl, & Niederhoffer 2002). ED comments also use more words expressing anger. This suggests that for people experiencing ED, post-mortem profiles are providing a space for venting frustration, in contrast to the more traditional funerary language employed by non-ED comments.

One might imagine that to detect individuals on SNS who are struggling to cope with the loss of a friend, sentiment analysis would be a sufficient approach. Our results suggest otherwise. Post-mortem spaces are sites of social performance in which individuals (distressed or not) utilize highly emotive language, making sentiment analysis alone insufficient. Our findings suggest, however, that comments expressing emotional distress can be identified through the way in which sentiment is expressed.

Linguistic features demonstrate promise in differentiating ED comments from the rest of the post-mortem comments. People expressing ED showed higher use of first person singular pronouns and lower use of second person pronouns. This indicates that, overall, comments containing declarations of ED portray a linguistic style indicative of self-focus and isolation, deviating from the socially aware linguistic style of other post-mortem comments. Meanwhile, the decreased use of second person pronouns in comments expressing ED can be a sign that, for these authors, the comment space serves more as an environment for conveying individual yearning or pain, rather than continuing bonds by addressing the deceased.

The increased use of past tense verbs in distressed comments may suggest that looking back and reflecting on the past may be a form of coping for those who are experiencing intense episodes of sorrow, post-mortem. In a previous study, Pennebaker, Mayne & Francis (1997) found that decreased use of past tense words is one of the predictors of perceived emotional distress for those who are bereaving the loss of a life partner. The fact that in this study we see those expressing ED using higher rates of past tense words may suggest that those in distress are in fact using this space to help with their coping. Finally, future tense use contributed negatively to Model 1. It is possible that during ED, people are less inclined to talk about the future and are more focused on their past or present experiences.

ED comments score higher on use of adverbs and negations. Use of adverbs, specifically intensifier adverbs, which dominate the adverb dictionary of LIWC, have been found to be indicators of “female-style” language, in previous research (Mulac *et al.* 1998). Use of conjunctions and negations in language are also potential signs of intimacy between the speaker and the addressee. All of these parts of speech show higher occurrences in ED comments suggesting that ED comments possess more signs of intimacy than non-ED comments, which use more funerary-styled language.

Conclusion & Future Work

In this paper, we reported on the development and implementation of a coding system for classifying post-mortem SNS comments as expressing emotional distress (ED) or not. Using this coding system, we then conducted a thorough linguistic analysis of comments posted on MySpace profiles belonging to deceased users to identify differences between ED and non-ED comments with respect to linguistic style and sentiment content. We noted that significant linguistic style differences exist between ED and non-ED comments.

To account for possible higher order interactions between variables, we constructed two binary logistic regression models, identifying linguistic style and content features that are strong predictors for a comment being ED. The regression models show that within the linguistic style features we considered, levels of negations, pronouns, and adverbs show potential to be strong markers for detecting emotional distress within the context of grieving in social media outlets.

The major contributions of this work are in defining a systematic way to identify and classifying content that exhibits grief-related emotional distress, as well as laying the groundwork for characterizing the nature of comments in post-mortem spaces based on language use. Moreover, the results from this work provide indications of a relationship between language and psychological well-being, as observed in natural public environments.

This work marks an initial step towards using content in SNSs to identify emotionally distressed at a large scale. As such, this could result in tools to classify social media content similar to NLP-based work on classifying suicide letters (Pestian *et al.* 2010).

Identifying users who are struggling with the loss of a loved one can also result in development of tools or programs to intervene. Such interventions are already beginning to appear. For example, Facebook recently added functionality that allows users to report content generated by their friends as potentially suicidal, resulting in an e-mail sent to the content author with support

resources. It is not hard to imagine that users experiencing ED following their friends' passing might also benefit from similar support systems. Moreover by identifying users experiencing ED, system designers can create web experiences sensitive to the emotional states of mourners.

This study also contributes to our understanding of the writing of the bereaved in social media. While most individuals author non-distressed content, often utilizing funerary-styled language, individuals expressing emotional distress write in ways indicative of self-focus and isolation, despite the public nature of the SNS. This may suggest that even while SNS can benefit disenfranchised griever by allowing them to participate in a space of communal mourning (Carroll & Landry 2010), the public nature of the SNS alone may be insufficient for distressed individuals whose language suggests they remain isolated.

As an initial step at analyzing the language in post-mortem spaces in social media, our study has limitations. First, the data we used for both the development of the coding system and the language analysis consisted of content posted to MySpace between 2006 and 2007. Analysis of similar but more recent content from additional sites such as Facebook is necessary. Second, we limited the analysis of language use to exploratory models. Future research should build on this work to develop and test predictive models, possibly on other sources of post-mortem content. Additionally, the coding system for classifying emotional distress is not comprehensive; it is limited to the content that people post. For example, the pain associated with losing a loved one may prevent some users from interacting with a post-mortem profile entirely. Finally, LIWC, while a frequently used tool for language analysis has its weaknesses. Because it is a dictionary-based tool, it fails to capture contextual differences, and certain subtleties of language such as sarcasm.

While it can be a sensitive topic, death and mourning are unavoidable parts of life. As such, studying grief in the context of SNS is valuable. As the adoption of social media continues to grow, grief and mourning will increasingly become important aspects of our social experiences online.

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